

## RESPONSE TO REFEREE REPORT

MANUSCRIPT JTNN-2400:

*A COMPLETE ARITHMETIC STRATIFICATION OF DEGREE-2 POLYNOMIAL  
CONTINUED FRACTIONS: THE  $\mathcal{F}(2, 4)$  BASE CASE*

PAPANOKECHI

### COVER LETTER TO THE HANDLING EDITOR

Prof. Boris Adamczewski  
Handling Editor  
*Journal de Théorie des Nombres de Bordeaux*

Dear Prof. Adamczewski,

Thank you, and please convey our thanks to the referee, for the careful reading of manuscript JTNN-2400, “A Complete Arithmetic Stratification of Degree-2 Polynomial Continued Fractions: The  $\mathcal{F}(2, 4)$  Base Case.”

The manuscript proves the Completeness Conjecture for the  $\mathcal{F}(2, 4)$  family of degree-2 polynomial continued fractions and identifies the 24 transcendental-stratum families as Möbius transforms of  $\pi$  of the form  $K = (a + b\pi)/(c + d\pi)$  with  $a, b, c, d \in \mathbb{Z}$ , all sharing the structural coefficient ratio  $a_2/b_1^2 = -2/9$  and exhibiting Apéry-like recurrence structure. The classification proceeds by enumerating all 531,441 degree-2 families with integer coefficients bounded by  $D = 4$  and applying a three-stage PSLQ-based pipeline at precision  $\text{dps} = 300\text{--}500$ , yielding the decomposition  $\text{Rat}(113,270) \sqcup \text{Log}(0) \sqcup \text{Alg}(0) \sqcup \text{Trans}(24) \sqcup \text{Des}(400,093) = 513,387$  of the convergent subspace.

In the present submission we attach a revised manuscript in which all referee comments have been addressed, together with a point-by-point response below and a change-summary table at the end of this document. Sections, line numbers, and theorem labels in this response refer to the revised manuscript.

We hope the revisions are now satisfactory for publication in *JTNN* and remain at the editor’s and the referee’s disposal for any further clarification.

Yours sincerely,

Papanokechi  
Independent researcher, Yokohama, Japan  
ORCID: 0009-0000-6192-8273  
<https://github.com/papanokechi>

---

### POINT-BY-POINT RESPONSE TO REFEREE COMMENTS

**Response to Referee Comment 1. Comment:** [PENDING REFEREE COMMENT]

**Response:** [PENDING RESPONSE]

**Manuscript change:** [PENDING; section/line refs]

---

*Date:* [PENDING; date of response].  
ORCID: 0009-0000-6192-8273.

**Response to Referee Comment 2. Comment:** [PENDING REFEREE COMMENT]

**Response:** [PENDING RESPONSE]

**Manuscript change:** [PENDING; section/line refs]

**Response to Referee Comment 3. Comment:** [PENDING REFEREE COMMENT]

**Response:** [PENDING RESPONSE]

**Manuscript change:** [PENDING; section/line refs]

**Response to Referee Comment 4. Comment:** [PENDING REFEREE COMMENT]

**Response:** [PENDING RESPONSE]

**Manuscript change:** [PENDING; section/line refs]

**Response to Referee Comment 5. Comment:** [PENDING REFEREE COMMENT]

**Response:** [PENDING RESPONSE]

**Manuscript change:** [PENDING; section/line refs]

---

#### SUMMARY OF CHANGES

| Comment# | Section           | Line range        | Change type       | Brief description |
|----------|-------------------|-------------------|-------------------|-------------------|
| 1        | [PENDING REFEREE] | [PENDING REFEREE] | [PENDING REFEREE] | [PENDING RE       |
| 2        | [PENDING REFEREE] | [PENDING REFEREE] | [PENDING REFEREE] | [PENDING RE       |
| 3        | [PENDING REFEREE] | [PENDING REFEREE] | [PENDING REFEREE] | [PENDING RE       |
| 4        | [PENDING REFEREE] | [PENDING REFEREE] | [PENDING REFEREE] | [PENDING RE       |
| 5        | [PENDING REFEREE] | [PENDING REFEREE] | [PENDING REFEREE] | [PENDING RE       |

---

#### AI DISCLOSURE

During the preparation of this work the author used GitHub Copilot (Microsoft) and Anthropic Claude in order to assist with code generation, numerical computations, and manuscript drafting. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of this article.

YOKOHAMA, JAPAN

URL: <https://github.com/papanokechi>